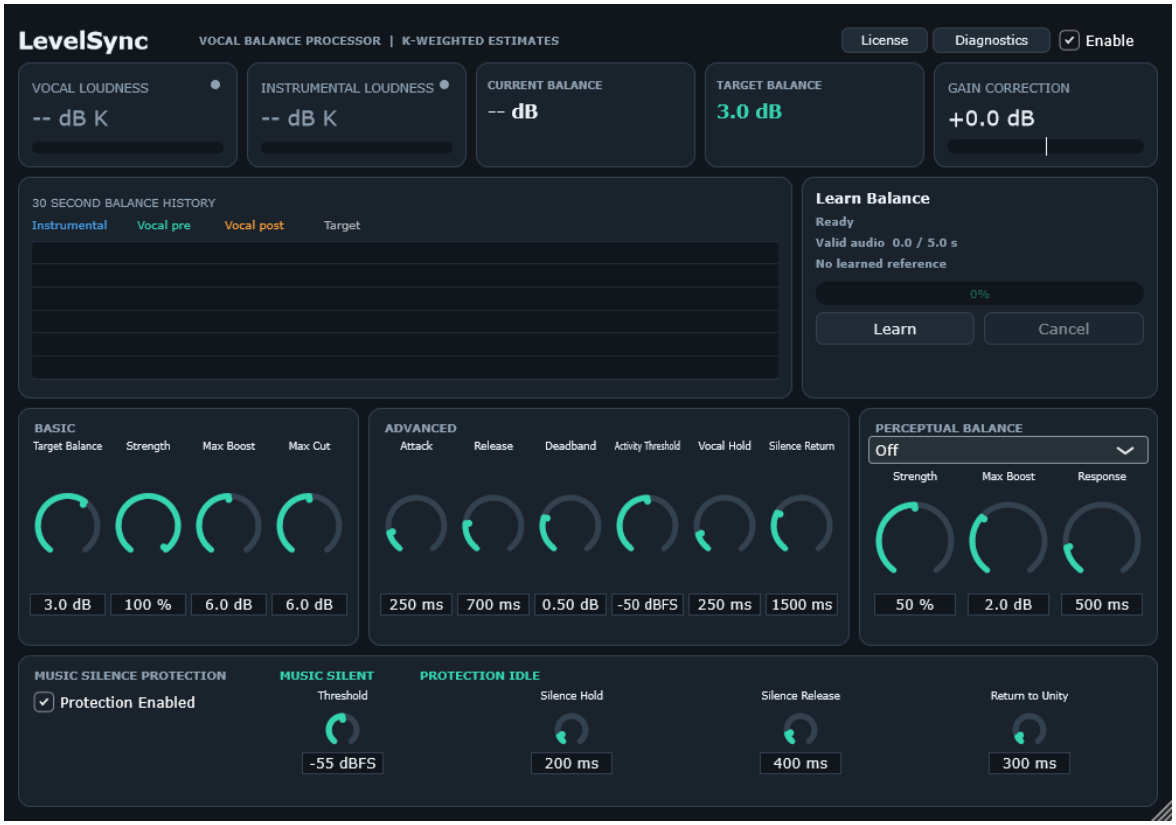
# LevelSync
# User Manual

Automatic vocal balancing for VST3 and Standalone




## Version 0.5.1 | Windows x64 | November 2026

This manual documents the interface and behavior implemented in LevelSync 0.5.1.

# 1 Overview and quick start

LevelSync smoothly adjusts vocal gain against a stereo instrumental sidechain. The main input carries vocals; the auxiliary bus is used only as a reference and is never mixed into the vocal output.

## Signal flow

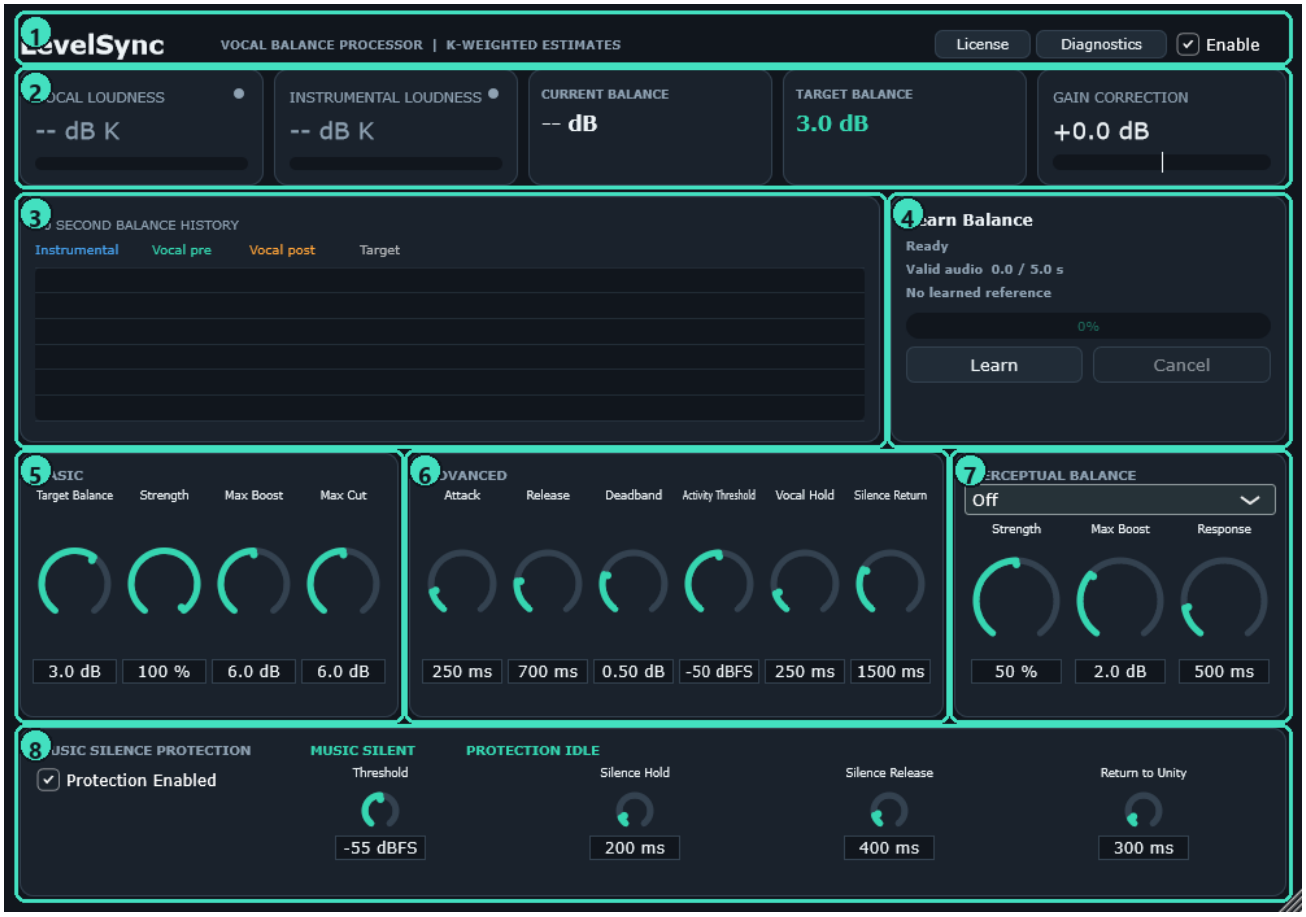
Vocal input → activity/loudness analysis → balance controller → gain smoothing → vocal output. The instrumental sidechain feeds only loudness, silence, and masking analysis.

## Quick start

- Insert LevelSync on the vocal track.
- Route the instrumental track or bus to the stereo sidechain.
- Play vocals and music together and confirm both meters are valid.
- Set Target Balance, gain limits, Attack, and Release, or run Learn Balance.
- Keep Music Silence Protection enabled for instrumental stops.

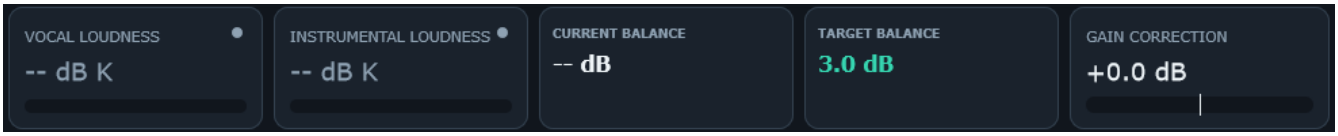
Displayed loudness values are 400 ms K-weighted estimates, not normative integrated or short-term LUFS measurements. Music silence detection uses a separate dBFS energy measurement.

# 2 Interface map



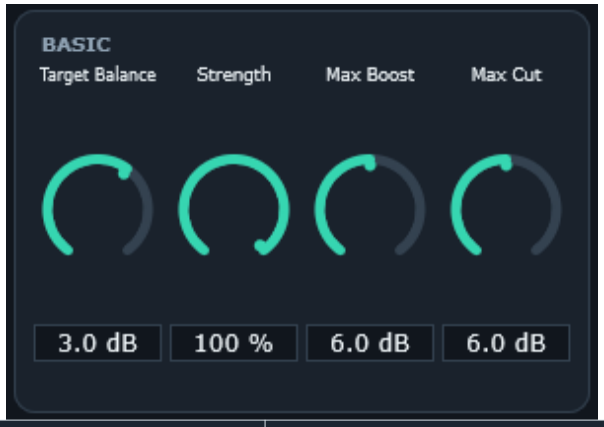

| # | Area | Purpose |
|---|---|---|
| 1 | Header | License, Diagnostics, and global Enable. |
| 2 | Monitoring | Vocal/instrumental loudness, current/target balance, and correction. |
| 3 | History | Last 30 seconds of instrumental, pre/post vocal, and target. |
| 4 | Learn Balance | Automatic vocal-to-instrumental calibration. |
| 5 | Basic | Target, strength, and boost/cut limits. |
| 6 | Advanced | Timing, deadband, vocal activity, and silence return. |
| 7 | Perceptual Balance | Optional masking-based correction. |
| 8 | Music Silence Protection | Vocal protection during instrumental stops. |

# 3 Monitoring and global commands

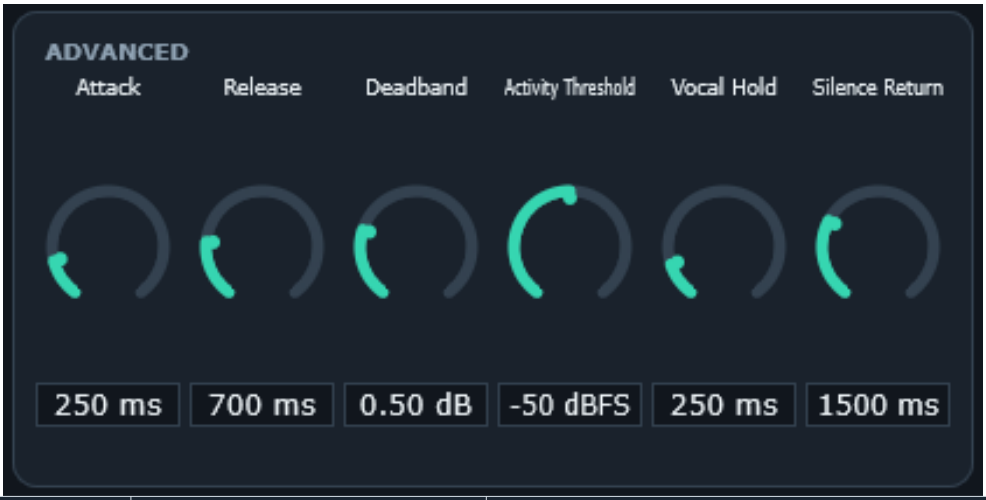



| Element | Meaning |
|---|---|
| Vocal Loudness | Pre-correction K-weighted vocal estimate. '-- dB K' means no valid measurement. |
| Instrumental Loudness | K-weighted sidechain estimate. Invalid sidechain keeps control neutral. |
| Current Balance | Current vocal/instrumental difference when both measurements are valid. |
| Target Balance | Desired relationship, set manually or by Learn Balance. |
| Gain Correction | Smoothed gain currently applied to the vocal signal. |
| Enable | Enables or disables balancing; disabled control returns safely to 0 dB. |
| Diagnostics | Shows algorithm telemetry and spectral analysis. |
| License | Shows AGPL and third-party notice information. |

# 4 Basic and Advanced controls

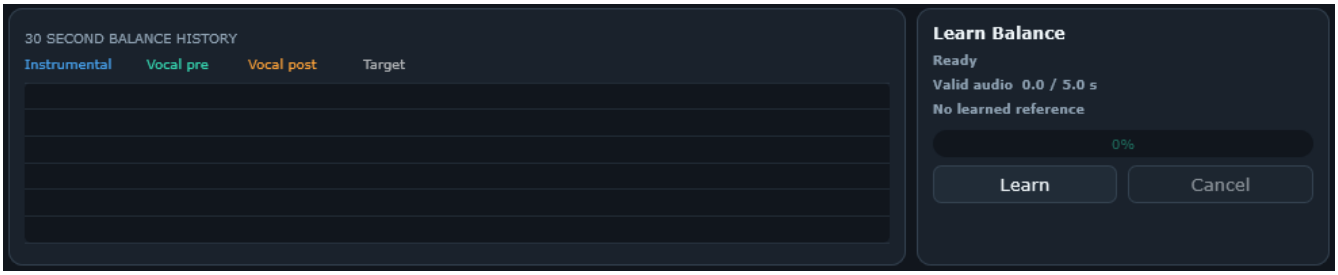



| Basic control | Range / Default | Function |
|---|---|---|
| Target Balance | -12..12 dB / 3 dB | Desired vocal position relative to the instrumental. |
| Strength | 0..100% / 100% | Scales the calculated correction. |
| Max Boost | 0..12 dB / 6 dB | Maximum positive vocal gain. |
| Max Cut | 0..12 dB / 6 dB | Maximum vocal attenuation. |




| Advanced control | Range / Default | Function |
|---|---|---|
| Attack | 50..1500 ms / 250 | Speed when reducing gain. |
| Release | 100..3000 ms / 700 | Speed when releasing correction. |
| Deadband | 0..2 dB / 0.5 | Neutral zone that prevents small movements. |
| Activity Threshold | -80..-20 dBFS / -50 | Vocal activity threshold. |
| Vocal Hold | 0..2000 ms / 250 | Holds activity between syllables. |
| Silence Return | 100..5000 ms / 1500 | Return to 0 dB without reliable vocal activity. |

# 5 Learn Balance




- Click Learn while the mix has a musically appropriate vocal/instrumental relationship.
- The plugin first returns gain smoothly to unity, then collects 5 seconds of valid audio.
- Progress advances only with reliable vocal activity and valid musical sidechain.
- Instrumental silence pauses calibration and preserves the previous reference.
- Completion updates Target Balance and stores the learned spectral profile.
- Cancel stops the current session without replacing the previous valid reference.

| State | Meaning |
|---|---|
| Ready | Available to start. |
| Preparing | Returning gain to unity. |
| Learning valid audio | Collecting reliable material. |
| Completed | Reference calculated and applied. |
| Failed | Not enough valid audio. |
| Cancelled | Stopped by the user. |

# 6 Perceptual Balance




Perceptual Balance analyzes vocal/instrumental competition in 24 ERB bands. It adds only positive correction and requires vocal activity, valid spectral measurement, and a learned reference.

| Control | Options / Range | Function |
|---|---|---|
| Mode | Off / Gentle / Balanced / Strong | Intensity scale: 0x, 0.5x, 1x, or 1.5x. |
| Strength | 0..100% / 50% | Scales perceptual correction. |
| Max Boost | 0..6 dB / 2 dB | Limits additional perceptual boost. |
| Response | 50..3000 ms / 500 ms | Smooths masking response. |

## Shared headroom

Perceptual correction uses only the headroom remaining under global Max Boost. It cannot push total gain beyond that limit.

# 7 Music Silence Protection




The protection prevents an old vocal reduction from remaining during an instrumental stop. It uses unweighted stereo energy in dBFS, separately from the K-weighted balance estimates.

| Control | Range / Default | Function |
|---|---|---|
| Protection Enabled | On | Enables protection. |
| Threshold | -90..-20 dBFS / -55 | Starts silence timing below this level. |
| Silence Hold | 0..2000 ms / 200 | Confirms silence after this duration. |
| Silence Release | 0..3000 ms / 400 | Confirms music after threshold + 3 dB. |
| Return to Unity | 50..3000 ms / 300 | Smoothly returns vocal gain to 0 dB. |

During confirmed silence, new instrumental-based corrections stop, Learn pauses, perceptual correction is discharged, and vocal gain returns to unity independently of vocal presence.

# 8 Diagnostics and history

Diagnostics displays algorithm telemetry for verification; it is not a normative loudness meter.

| Reading | Interpretation |
|---|---|
| Vocal pre / Music | K-weighted estimates before correction. |
| Current / Target | Measured and desired balance. |
| Error / Target gain | Error outside deadband and requested gain. |
| Applied | Gain after smoothing. |
| Vocal active | Vocal activity detector state. |
| Sidechain valid | Reliable instrumental measurement availability. |
| Max Boost / Max Cut | Whether the target reached a limit. |
| Masking / confidence | Spectral competition and analysis confidence. |
| Perceptual | Additional perceptual boost in dB. |

## 30-second history

- Instrumental: sidechain level.
- Vocal pre: vocal level before correction.
- Vocal post: vocal level after applied gain.
- Target: desired vocal level derived from instrumental level and Target Balance.

Missing lines indicate invalid measurements, not invented zero values. UI telemetry does not participate in audio control.

# 9 Installation and routing

## VST3 installation

- Close LevelSync instances in the DAW.
- Use the installer or copy the complete LevelSync.vst3 folder to a directory scanned by the DAW.
- The standard Windows directory is C:\Program Files\Common Files\VST3.
- Rescan VST3 plugins and confirm detection.

## DAW routing

- Insert LevelSync on the mono or stereo vocal track.
- Send the instrumental track or bus to the stereo auxiliary input.
- Do not route vocal output back into the sidechain.
- Confirm valid Instrumental Loudness before evaluating balance.

## Standalone

Standalone uses the same AudioProcessor and editor as VST3. Because its wrapper has no separate instrumental sidechain, automatic control remains neutral. Use it to inspect the interface and configure audio devices, not to simulate real balancing.

# 10 Recommended settings and troubleshooting

| Situation | Suggested adjustment |
|---|---|
| Vocal pumps or breathes | Increase Release, Deadband, or Vocal Hold. |
| Slow reaction to strong entries | Reduce Attack gradually. |
| Vocal rises too much | Reduce Max Boost and preserve headroom. |
| Noise activates control | Raise Activity Threshold above the noise floor. |
| Protection chatters | Increase Silence Hold/Release or adjust Threshold. |
| Learn does not advance | Confirm active vocals and valid musical sidechain. |
| No perceptual correction | Run Learn, enable a mode, and confirm valid signals. |

## Common issues

- Plugin missing: confirm x64, scanned VST3 folder, and rescan.
- Instrumental Loudness shows '--': check the stereo sidechain route and DAW auxiliary-bus support.
- Standalone does not balance: expected without a separate sidechain.
- SmartScreen warning: public builds are unsigned; verify the published SHA-256.

# 11 Complete parameter reference

| Parameter | Unit | Minimum | Maximum | Default |
|---|---|---|---|---|
| Balance Offset | dB | -12 | 12 | 3 |
| Strength | % | 0 | 100 | 100 |
| Max Boost | dB | 0 | 12 | 6 |
| Max Cut | dB | 0 | 12 | 6 |
| Attack | ms | 50 | 1500 | 250 |
| Release | ms | 100 | 3000 | 700 |
| Deadband | dB | 0 | 2 | 0.5 |
| Vocal Threshold | dBFS | -80 | -20 | -50 |
| Vocal Hold | ms | 0 | 2000 | 250 |
| Silence Return | ms | 100 | 5000 | 1500 |
| Perceptual Mode | - | Off | Strong | Off |
| Perceptual Strength | % | 0 | 100 | 50 |
| Max Perceptual Boost | dB | 0 | 6 | 2 |
| Masking Response | ms | 50 | 3000 | 500 |
| Music Silence Threshold | dBFS | -90 | -20 | -55 |
| Music Silence Hold | ms | 0 | 2000 | 200 |
| Music Silence Release | ms | 0 | 3000 | 400 |
| Music Return to Unity | ms | 50 | 3000 | 300 |
| Music Silence Protection | bool | Off | On | On |
| Enable | bool | Off | On | On |

## Automation and state

Parameters are exposed through AudioProcessorValueTreeState and persist in DAW projects. The learned reference and spectral profile are serialized. Automation is read once per block, not sample-accurately.

# 12 Limitations, safety, and licensing

- LevelSync does not separate sources; bleed, noise, and reverberation can affect detectors.
- Positive gain can exceed 0 dBFS; preserve headroom and use appropriate downstream metering/limiting.
- VST3 compatibility does not guarantee identical sidechain routing in every DAW.
- The sidechain is read-only and is never mixed into vocal output.
- Validate with your DAW version, driver, sample rate, and buffer.

## Licensing

Original LevelSync code is distributed under GNU Affero General Public License v3.0 or later. The project uses JUCE 9.0.5 under the AGPL option. See LICENSE, NOTICE, and THIRD_PARTY_NOTICES.md. The software is provided without warranty.

## Downloads and source

Releases: https://github.com/ricferreiradev/levelsync/releases
Source: https://github.com/ricferreiradev/levelsync

## Useful reports

Include LevelSync and DAW versions, bus layout, sample rate, buffer, driver, reproduction steps, and a Diagnostics screenshot when possible. Review confidential content before sharing projects or audio.

## LevelSync v0.5.1

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